Computer

Class 8 (English Medium)

Solved Exercises — Sindh Textbook Board

6 lessons — 126 questions

آئي سي ٹي جا بنيادي اصول

[1-1] How does 5G technology improve network connectivity?

(a) With slow speed

(b) With less bandwidth

✓ (c) With fast speed and low latency

(d) With fewer connected devices

[1-2] What does IoT (Internet of Things) enable devices to do?

(a) Work separately

✓ (b) Connect with other devices and share data

(c) Work only on their own

(d) Run without the internet

[1-3] What is the main purpose of edge computing?

(a) Sending data to a faraway server

✓ (b) Processing data where it is produced

(c) Slowing down the network

(d) Not storing data

[1-4] What does a router do in a network?

(a) Connect devices inside a local network

✓ (b) Connect different networks and deliver data to the correct destination

(c) Increase internet speed

(d) Connect wireless devices to the network

[1-5] What is the main job of a firewall in network security?

(a) Blocking access to the internet

✓ (b) Monitoring and controlling incoming and outgoing network traffic

(c) Stopping data from being shared

(d) Improving device speed

[1-6] What is a feature of IoT devices in a network?

(a) They stay isolated from other devices

✓ (b) They stay connected with other devices and share data

(c) They need no operating system

(d) They run without the internet

[1-7] What happens as a result of faults in a network?

(a) Speed increases

✓ (b) Data is lost or damaged

(c) Better communication between devices

(d) Easier access to resources

[1-8] What does edge computing help to reduce in a network?

✓ (a) Latency

(b) Device connectivity

(c) Bandwidth use

(d) Internet access

[1-9] How does 5G help IoT?

✓ (a) By connecting more devices

(b) By separating networks

(c) By reducing speed

(d) By moving devices

[1-10] Through what do devices in a network communicate with each other most effectively?

(a) Hub

(b) Router

✓ (c) Switch

(d) Modem

Question: [1-11] Fill in the blank: ____ makes high-speed connectivity

possible for different devices and gives better connectivity.

Answer: 5G — the 5G network provides high-speed connectivity and better connectivity for different devices.

Question: [1-12] Fill in the blank: ____ is the process in which data is processed near the place where it is produced, which reduces latency and bandwidth use.

Answer: Edge computing — processing data near its source is edge computing.

Question: [1-13] Fill in the blank: A ____ is a device that connects different networks and makes sure data reaches the correct destination inside the network.

Answer: Router — a router connects different networks and decides the path for data.

Question: [1-14] Fill in the blank: In a network, a ____ guides data between connected devices, which makes communication in a local area network (LAN) effective.

Answer: Switch — a switch sends data to the correct device, which makes communication in a LAN effective.

Question: [1-15] Fill in the blank: ____ are devices that collect data from their surroundings and pass it to IoT, which enables automatic processes in smart homes and other industries.

Answer: Sensors — sensors collect data from the surroundings and pass it to the IoT system.

Question: [1-16] Fill in the blank: ____ is the technology that connects physical things to the internet, so that they can send and receive data and stay connected for better communication.

Answer: IoT (Internet of Things) — in IoT, physical things connect to the internet and send and receive data.

Question: [1-17] Fill in the blank: In a network, the main job of a ____ is to monitor and control incoming and outgoing traffic, so that unauthorised access can be prevented.

Answer: Firewall — a firewall checks traffic against rules and blocks unauthorised access.

Question: [1-18] Fill in the blank: ____ are systems fitted inside devices such as machines, which complete tasks automatically and make sure they run without human intervention.

Answer: Embedded systems — an embedded system is a small computer fitted inside a bigger device that performs a particular job.

Question: [1-19] Fill in the blank: ____ is the technology that improves connectivity for more devices, so that self-driving cars and smart cities get a strong connection.

Answer: 5G — 5G connects more devices and supports uses such as self-driving cars and smart cities.

Question: [1-20] Fill in the blank: A ____ is the device that organises communication between devices in a wide area network (WAN) and joins LANs over long distances.

Answer: Router — in a WAN, a router organises communication between different LANs and delivers data to its destination.

Question: [1-21] What are the main components of a computer network, and how do they make effective communication possible?

Answer: The main components of a network are the sender, the message, the medium, the receiver, and the protocol. The sender sends the message, the message is the information being sent, the medium carries the message, the receiver gets the message, and the protocol is the set of rules that lets both sides connect properly. In addition, hardware such as the NIC, modem, hub, switch, router, gateway, and access point lets

devices join the network.

Question: [1-22] What challenges does network security face, and what measures can be taken to protect information?

Answer: The challenges are: unauthorised access, faults that can lose or damage data, and viruses and malware that can spread through the network. For protection, use a firewall, strong passwords, two-factor authentication, up-to-date antivirus software, and secure communications.

Question: [1-23] How does IoT improve daily life, and can you give an example from health?

Answer: IoT makes life easier by connecting things to the internet, with automatic control and better convenience. A health example: medical devices such as MRI machines rely on embedded systems to work correctly, and connected health devices help in monitoring patients.

Question: [1-24] What is the role of sensors in IoT devices, and how do they make remote control possible?

Answer: Sensors collect data from the surroundings, such as temperature or movement, and send it to the IoT system. On the basis of that data, devices work automatically, or the user can control them from far away through a phone, such as setting the home temperature from a phone.

Question: [1-25] What is an embedded system, and how does it make devices such as washing machines automatic?

Answer: An embedded system is a small computer fitted inside a bigger device that performs a particular job. In a washing machine it sets the washing time, controls the temperature, and runs the wash cycle automatically, so the machine works without human intervention.

Question: [1-26] Name three devices around you that use embedded systems and explain their function.

Answer: (1) Car — an embedded system controls systems such as the

engine and airbags. (2) Home appliances such as a washing machine and refrigerator — they set time and temperature and work automatically. (3) Medical devices such as an MRI machine — an embedded system makes sure it works correctly.

Question: [1-27] What is edge computing, and how does it reduce delay in data processing?

Answer: In edge computing, data is processed near the place where it is produced, at the device or sensor itself. The data does not need to travel to a faraway central server, so travel time is reduced and the response comes quickly.

Question: [1-28] How can edge computing be used in a traffic management system?

Answer: In a traffic system, edge computing processes data from road sensors on the spot, for example changing signal timings by watching traffic flow or telling drivers a better route, so that decisions are made quickly and at the right time.

Question: [1-29] How does 5G technology improve connectivity, and which industries benefit most from it?

Answer: 5G gives better connectivity, faster speed, low latency, and support for many devices at the same time. The greatest benefit goes to fields that need quick responses and many devices, such as self-driving cars, smart cities, health, and industry.

Question: [1-30] What are the possible risks that come with the arrival of 5G, and how can they be reduced?

Answer: Possible risks are: a greater risk of unauthorised access as more devices get connected, and worries about data protection. To reduce them, use a firewall, strong passwords, two-factor authentication, and up-to-date security software, and follow network security rules.

ڈجیٹل مہارتون

[2-1] What is Microsoft Excel?

✓ (a) Spreadsheet

(b) Database management

(c) Presentation

(d) Workbook

[2-2] What is the intersection of a column and a row in a worksheet called?

(a) Column

(b) Value

✓ (c) Cell

(d) Address

[2-3] With what do Excel formulas begin?

(a) /

(b) []

✓ (c) =

(d) &

[2-4] Which Excel feature shows only the data that meets a particular condition?

(a) Sorting

(b) Formula

✓ (c) Filtering

(d) Pivot

[2-5] Where is conditional formatting in Excel?

(a) Insert tab

(b) Page Layout tab

(c) Data tab

(d) Home tab

[2-6] Which Excel function checks whether a condition is true?

Question: [2-7] Write the uses of Microsoft Excel.

Answer: Personal use: keeping track of your budget, contact lists, and expenses. School: organising homework and assignments, recording grades, and making charts. Work: analysing sales data, making employee schedules, and tracking finances.

Question: [2-8] What is the purpose of the formula bar?

Answer: The formula bar shows the content or formula of the active cell and lets you type or change it. For example, when cell H6 is selected, the formula bar shows =C6+D6+E6+F6+G6.

Question: [2-9] What is the difference between a workbook and a worksheet?

Answer: A workbook is a complete file, like a binder, containing one or more worksheets, saved in .xlsx format. A worksheet is a single page made of a grid of rows and columns, where data is entered and organised.

Question: [2-10] How do you insert a chart in a worksheet? Write the steps.

Answer: (1) Select the data to show in the chart, for example students' names and their marks. (2) Click the Insert tab. (3) Click the Chart command. (4) Choose the chart type you want, for example a 3D bar chart — the chart will appear in the worksheet.

Question: [2-11] Activity: prepare a simple sales report using the Home tab options.

Answer: (1) Open Excel and enter the data: Product, Quantity, Price, Total; Apple 10 / 2.5, Banana 15 / 1.8, Orange 8 / 3. (2) Make the headings Bold (B) from the Font group and centre them with the Alignment group. (3) Give the headings a fill colour. (4) Apply currency format to Price and Total



from the Number group. (5) Apply borders to the whole table from the Font group. (6) For Total, type the formula =B2*C2 and fill it down: Apple 25.00, Banana 27.00, Orange 24.00. (7) Save the file as BasicActivity.xlsx.

Question: [2-12] Activity: make a daily expenses calculator.

Answer: (1) Open a new workbook and type the headings: Date, Category, Description, Amount. (2) Enter the entries: 11/1/2024 Food Grocery shopping 50; 11/2/2024 Transport Bus fare 15; 11/3/2024 Rent Monthly rent payment 500; 11/4/2024 Transport Train fare 10. (3) Change the Date column to long date with number format (Ctrl+1) and apply currency format to Amount. (4) For the total expense, type =SUM(D2:D8) in D9 — the result is $575. (5) For each category total, use SUMIF: Food =SUMIF(B2:B8,"Food",D2:D8) = 50; Transport =SUMIF(B2:B8,"Transport",D2:D8) = 25; Rent =SUMIF(B2:B8,"Rent",D2:D8) = 500. (6) Save the file as Personal Expense.xlsx.

Question: [2-13] Task: in the movies worksheet, work out profit, highest/lowest/average gross, sort the data, and make a pie chart.

Answer: The movies are in rows 2 to 12, Budget in column C and World Gross in column D. Profit (column E): =D2-C2, that is, subtract the budget from the gross, and fill it down. Highest gross: =MAX(D2:D12). Lowest gross: =MIN(D2:D12). Average gross: =AVERAGE(D2:D12). Sorting: select the data, use Sort from the Data tab, choose Profit in Sort by, and choose Largest to Smallest. Pie chart: select Movie and Profit, then choose a Pie chart from the Insert tab.

Question: [2-14] Task: prepare any one of the following worksheets and also make a chart: student mark sheet, monthly bill, attendance register.

Answer: For example, a mark sheet: write the columns — student name, registration number, subject marks, total, average, grade. For total use

=SUM(C6:G6), for average use =AVERAGE(C6:G6), for grade use the IF function, such as =IF(H6>=80,"A",IF(H6>=60,"B","C")). Then select names and totals and make a bar chart from the Insert tab. In a monthly bill, write date, item, quantity, and amount, and use SUM for the total; in an attendance register, write student, dates, and present/absent, and work out each student's attendance percentage.

Question: [2-15] Task: in the toy shop worksheet, work out cost, discount, final cost, and totals.

Answer: Cost (column D): =B2*C2, that is, price per item times quantity ordered, and fill it down. Discount (column E): =D2*7%, that is, 7 percent discount on the cost, and fill it down. Final Cost (column F): =D2-E2, that is, subtract the discount from the cost, and fill it down. Total of all per-item prices (B7): =SUM(B2:B6). Grand totals: =SUM(D2:D6), total discount: =SUM(E2:E6), total final cost: =SUM(F2:F6).

Algorithmic Thinking and Problem Solving

Question: [3-1] Activity: look at the four problems below and decide whether each one is simple or complex, and write the reason.

Answer: (1) The teacher has 5 apples, the friend has 3 more — how many apples does the friend have? Simple — just one step: 5 + 3 = 8. (2) Organising a class trip: bus booking, a whole-day schedule, activities for different groups — Complex — many steps and many factors come together. (3) Ali is making tea by a recipe and has to make it for many people at a party — Simple — the steps are fixed, only the quantities have to be increased in proportion. (4) A science exhibition in school: getting permission, making a schedule, deciding who will present what — Complex — many steps and many decisions come together.

Question: [3-2] Activity: you have to arrange a pizza party for 20 students of your class. Each student needs 2 slices of pizza, each pizza has 8 slices, and one pizza costs 1000 rupees. How many pizzas must be bought, and what will the total cost be? Write the steps.

Answer: Steps: (1) Start. (2) Slices needed = 20 × 2 = 40. (3) Pizzas needed = 40 ÷ 8 = 5. (4) Total cost = 5 × 1000 = 5000 rupees. (5) Show the result: 5 pizzas, 5000 rupees. (6) End.

Question: [3-3] Activity: write pseudocode to find the area of a circle.

Answer: Start // pseudocode for the area of a circle; Input radius; Area = 3.14 × radius × radius; Display Area; End.

Question: [3-4] Activity: write pseudocode to convert Celsius to Fahrenheit.

Answer: Start // converting Celsius to Fahrenheit; Input Celsius; Fahrenheit = (Celsius × 9 ÷ 5) + 32; Display Fahrenheit; End.

Question: [3-5] Activity: identify the variables and constants in the juice

shop situation.

Answer: Variables (values that change): the number of juice glasses sold, and the total money earned. Constants (values that do not change): the price of each glass, 100 rupees, and the shop's opening time, 1:00 to 3:00 pm.

Question: [3-6] Activity: which type of conditional statement is used to describe the following scenarios? Write the name of the statement.

Answer: (1) If a student's marks are 60, he passes — IF. (2) If Umair saved 100 rupees from his pocket money, he can buy ice cream, otherwise he must save more — IF-ELSE. (3) A student can go on the field trip if the permission slip is signed and the fee is paid, otherwise show a message — IF-ELSE. (4) Clothes by weather: below 15°C wear warm clothes, above 25°C wear a light shirt, above 35°C wear shorts, otherwise check the weather — IF-ELIF-ELSE. (5) If it is raining, take an umbrella — IF. (6) Hassan first checks whether it is the weekend, then checks whether his homework is complete — Nested IF. (7) Sohaib decides whether he has enough money to buy ice cream — IF-ELSE. (8) Samia decides whether she can go out late at night, only with permission and when it is the weekend or the next day is not a school day — IF-ELSE.

[3-7] What is an algorithm?

(a) A type of computer

✓ (b) A set of instructions to solve a problem

(c) A programming language

(d) A game

[3-8] _____ is an example of a conditional statement.

(a) Loop

✓ (b) If-Else

(c) Print

(d) Variable

[3-9] In algorithmic thinking, the purpose of breaking a big problem into smaller problems is _____.

(a) Making the problem more difficult

✓ (b) Solving the problem more easily

(c) Adding more steps

(d) Avoiding the computer

[3-10] A computer uses _____ to make a decision.

(a) Loop

(b) Variable

✓ (c) Conditional Statements

(d) Constant

[3-11] Identifying patterns, relations, and similarities in data and using them in a problem is called _____.

(a) Abstraction

(b) Decomposition

✓ (c) Pattern Recognition

(d) Algorithmic Thinking

[3-12] Focusing only on important details and ignoring unneeded information is called _____.

(a) Decomposition

✓ (b) Abstraction

(c) Pattern Recognition

(d) Algorithm Design

[3-13] An example of a nested loop is _____.

(a) A loop that runs once

✓ (b) A loop inside another loop

(c) A loop that never runs

(d) An IF-ELSE-IF condition

[3-14] In IF-ELIF-ELSE, when does the Else part run?

(a) When all conditions are true

✓ (b) When all conditions are false

(c) When a repeat loop is used

(d) When a nested loop is used

[3-15] _____ in a problem lets us break a complex problem into smaller parts and prepare an algorithm to solve it.

✓ (a) Computational Thinking

(b) Nested Loop

(c) Conditional Statements

(d) Pseudocode

[3-16] An example of a "for loop" is _____.

(a) "Set x = 5"

✓ (b) "Repeat 10 times"

(c) "Print a message"

(d) "Set a variable"

Question: [3-17] What are variables and constants in pseudocode? Why are they used?

Answer: A variable is a named container whose value can change during a program, such as score or student_name. A constant is a named container whose value does not change during a program, such as Pi = 3.14, hours in a day = 24, or days in a week = 7. They are used to store data, to make programs easy to read, and to avoid writing the same value again and again.

Question: [3-18] What is the difference between sequential flow and conditional flow?

Answer: In sequential flow, the steps of a program run one after another,

in order. In conditional flow, a step runs only when some condition is true, as in IF or IF-ELSE statements — different paths are taken according to the result of the condition.

Question: [3-19] What does decomposition do in problem solving? Give an example.

Answer: Decomposition splits a big problem into small, manageable parts, so that each part can be solved separately. Example: to build a LEGO car, split the problem into parts — making the chassis, fitting the wheels, making the body, fitting the roof, and final decoration.

Question: [3-20] How does abstraction help in solving a problem? Give an example.

Answer: In abstraction, attention is given only to important details and unneeded details are left out, which makes the problem simple and clear. Example: when building a LEGO car, the car's colour, the design of the seats, or the details of the headlights are ignored, and attention is given only to the main structure.

Question: [3-21] What is the halting problem, and why can some problems not be solved by a computer?

Answer: The halting problem is the question of whether a program can tell if another program will stop or run forever. Some problems cannot be solved by a computer, such as a program stuck in an infinite loop that never stops and keeps the CPU busy.

Question: [3-22] Write pseudocode: read three numbers and display their sum and average.

Answer: Start; Input first number; Input second number; Input third number; Sum = first + second + third; Average = Sum ÷ 3; Display Sum; Display Average; End.

Question: [3-23] Write pseudocode: count the numbers from 1 to 100.

Answer: Start; Set count = 1; While count <= 100: Display count; count = count + 1; End.

Question: [3-24] Write an algorithm: check whether a given number is even or odd.

Answer: Start; Input the number; If the remainder of number ÷ 2 is 0, display "It is even", otherwise display "It is odd"; End.

Question: [3-25] Write an algorithm: find the largest of three numbers.

Answer: Start; Input three numbers: a, b, c; If a >= b and a >= c, display a; Else if b >= a and b >= c, display b; Else display c; End.

Question: [3-26] Class activity: think of a problem that can be solved by different methods. Write an algorithm for all the methods and tell which solution is best and why.

Answer: Problem: making tea. Method 1 (kettle on stove): put water in the kettle, heat it on the stove, pour it into a cup, add tea leaves and sugar. Method 2 (electric kettle): heat water in the electric kettle, pour it into a cup, add tea leaves and sugar. The best solution is Method 2, because the electric kettle heats water quickly and needs less watching — the same result comes in less time.

Question: [3-27] Class activity: write an algorithm that decides whether a student qualifies for a scholarship — on the basis of grade.

Answer: Start; Input the student's grade; If grade is A+, display "100% scholarship"; Else if grade is A, display "80% scholarship"; Else if grade is B, display "50% scholarship"; Else display "No scholarship"; End.

Question: [3-28] Write an algorithm: work out the total sales of a shop for each day, assuming there are 30 days in the month.

Answer: Start; Set total = 0; Set day = 1; While day <= 30: Input that day's sales; total = total + that day's sales; day = day + 1; Display total; End.

Question: [3-29] Write an algorithm: convert given inches into centimetres.

Answer: Start; Input inches; Centimetres = inches × 2.54; Display centimetres; End.

پروگرامنگ (اسکریچ)

Question: [4-1] Activity: make a Scratch program that identifies whether a given number is positive or not. Write the algorithm for its decision.

Answer: Program: when the flag is clicked, set the Number variable to 0, ask "What's your number?", and store the answer in Number. Then check: if Number > 0, say "It is positive", otherwise say "It is not positive". Algorithm: Start; Input the number; If number > 0, display "It is positive", otherwise display "It is not positive"; End.

Question: [4-2] Make a Scratch program that checks whether a given number is even or odd.

Answer: When the flag is clicked: set Number to 0; ask "What's your number?" and wait; set Number to answer. Then If-Then-Else: if Number mod 2 = 0, say "The Number is Even" for 2 seconds, otherwise say "The Number is Odd" for 2 seconds. Example: entering 15 gives "The Number is Odd".

Question: [4-3] Make a Scratch program with a nested conditional: check whether the number is positive or not, and if it is positive, whether it is even or odd.

Answer: Outer condition: if Number > 0, then the inner If-Then-Else: if Number mod 2 = 0, say "The Number is Even", otherwise say "The Number is Odd". Outer Else: say "The number is negative or zero". Example: 15 gives "The Number is Odd", -2 gives "The number is negative or zero".

Question: [4-4] Make three types of loop programs in Scratch: Repeat, Repeat Until, and Forever.

Answer: (1) Repeat 10: when the flag is clicked, point in direction 90,

repeat 10 times: next costume, move 10 steps, wait 1 second, turn 15 degrees. (2) Repeat Until: set counter to 0, repeat until counter = 10: change counter by 1, next costume, move 10 steps, wait 1 second, turn 15 degrees. (3) Forever: forever: wait 1 second, say "Meow", start sound Meow.

Question: [4-5] Make a nested loop program in Scratch: move the sprite and change the backdrop.

Answer: (1) Choose a backdrop from Choose a Backdrop and choose a sprite. (2) Place the sprite at the starting position with the go to x: -145 y: -77 block. (3) Put a Repeat Until loop inside an outer Repeat 10 loop: repeat until counter = 4: change counter by 1, next costume, move 10 steps, wait 0.1 seconds. (4) After the Repeat Until loop, put a switch backdrop to next backdrop block so the backdrop changes after each round.

Question: [4-6] Make your own function (Custom Block) in Scratch: Calculate Area, to find the area of a rectangle.

Answer: From My Blocks, click Make a Block, name it "Calculate Area", and add two inputs: length and width. In the definition: set area to length × width. To use it: when the flag is clicked: ask length and store the answer in length, ask width and store the answer in width, run Calculate Area length width, then say area.

Question: [4-7] Cloning in Scratch: make clones of balloons.

Answer: Original sprite (balloon): when the flag is clicked, hide, forever: go to a random position (go to pick random -250 to 250, -200), next costume, create clone of myself, wait 0.5 seconds. Clone: when I start as a clone, show, repeat 50 times: point in a random direction from 80 to 100, change y by 10, change size by -2, wait 2 seconds; then delete this clone.

Question: [4-8] Make a simple maze game in Scratch.

Answer: (1) Make a maze backdrop (walls in blue colour) and add a sprite. (2) Movement: when the flag is clicked, go to x: -205 y: 147; four "when key pressed" blocks: up arrow (direction 0, 10 steps), down arrow (direction 180, 10 steps), right arrow (direction 90, 10 steps), left arrow (direction -90, 10 steps). (3) Hitting a wall: forever: if touching color blue, move 10 steps back. (4) Goal: if touching Ball, say "You win!". (5) Run the game and check it.

Question: [4-9] Make an advanced maze game in Scratch: several mazes, reaching the apple, next maze.

Answer: (1) Set up the backdrop and sprites: a maze backdrop, a sprite, and an Apple sprite as the goal. (2) Movement: four arrow-key blocks (up/down/right/left), in each: move 4 steps, then a nested check: if touching maze, move 4 steps back, otherwise if touching mouse-pointer, move 4 steps back. (3) Several mazes: when the flag is clicked, set size 10%, go to x: 10 y: -170, erase all, forever: five if blocks (one for each maze). (4) Goal: if touching Apple, say "You Win!", bring the sprite back to x: 10 y: -170, broadcast next maze, erase all. (5) When I receive next maze: next costume. (6) Run the game.

[4-10] What are the building blocks of all programming languages?



(a) Sequence

(b) Condition

(c) Loop

(d) Variable

[4-11] What is a loop inside another loop called?



(a) Nested

(b) If

(c) Iteration

(d) Cloning

[4-12] A loop that never runs has a condition that is _____.

(a) True

(b) False ✓

(c) Empty

(d) Zero

[4-13] The floor() function rounds a number _____.

(a) Down ✓

(b) Up

(c) To the nearest side

(d) To the other side

[4-14] Which function returns the largest value in a set of numbers?

Question: [4-15] What is a loop in Scratch, and how many types are there?

Answer: A loop is a block that makes an action repeat again and again, without writing it again. There are three types of loops in Scratch: (1) Repeat — repeats a fixed number of times. (2) Repeat Until — repeats until a condition becomes true. (3) Forever — never stops, repeats always.

Question: [4-16] What is the difference between if and if-else?

Answer: An if statement acts only when the condition is true; when the condition is false, it does nothing. An if-else statement has two paths: one action when the condition is true, and a different action when it is false.

Question: [4-17] Write some built-in functions of Scratch with examples.

Answer: pick random 1 to 10 — gives a random number between 1 and 10. abs of -3 = 3 — gives the absolute value of a number. join apple banana = "applebanana" — joins two words. round 45.5 = 46 — rounds to the nearest whole number. floor of 45.8 = 45 — rounds down. ceiling of 45.2 = 46 — rounds up. sqrt() — gives the square root of a number. max() — gives the largest value among numbers.

Question: [4-18] Teacher activities: (1) Make programs with patterns and

nested loops. (2) Compare the three types of loops. (3) Draw a square with a Repeat loop. (4) Make a game with a Repeat Until loop.

Answer: (1) Divide students into groups of 2-3; each group makes a program with patterns and nested loops, for example an outer loop that changes the backdrop and an inner loop that moves the sprite. (2) Each group does the same task with the three loops (Repeat, Repeat Until, Forever) and tells which loop suits which task. (3) Square: Repeat 4 [move 100 steps, turn 90 degrees]. (4) Game: Repeat Until touching (goal) [move the sprite with the arrows], then say "You win!".

Digital Citizenship

[5-1] What is digital citizenship?

(a) Using technology in business

✓ (b) Responsible use of technology and online behaviour

(c) Online games

(d) Watching videos online

[5-2] Which of the following is a positive online behaviour?

(a) Spreading false news

✓ (b) Staying connected with friends through social media

(c) Hacking others' accounts

(d) Sharing stolen material

[5-3] What is one health danger of excessive internet use?

(a) Better sleep

✓ (b) Anxiety and depression

(c) Better physical fitness

(d) More physical activity

[5-4] What is phishing?

(a) A type of fishing

✓ (b) A way of stealing personal data through fake websites

(c) A social media platform

(d) An online shopping tool

[5-5] Which is a way to keep information secure?

(a) Using weak passwords

(b) Sharing passwords with friends

(d) Sharing personal information on sites

[5-6] Fill in the blanks: (i) Digital citizenship guides us to adopt _____

behaviour online.

Answer: responsible

[5-7] Fill in the blanks: (ii) _____ allows online users to stay connected and share information.

Answer: internet

[5-8] Fill in the blanks: (iii) Cybercrime includes illegal activities carried out using _____.

Answer: computers and the internet.

[5-9] Fill in the blanks: (iv) One way to keep your online information safe is to use a _____ password.

Answer: strong

[5-10] Fill in the blanks: (v) Presenting someone else's work as your own without giving credit is called _____.

Answer: plagiarism.

Question: [5-11] What is digital citizenship?

Answer: Digital citizenship means using technology responsibly — keeping good behaviour online, respecting others, protecting personal information, and following the law.

Question: [5-12] What are the common uses of the internet?

Answer: Online shopping, communication (email, chat), marketing and brand awareness, customer service, advertising, social networking, getting news and information, and reading/listening to online material (news, magazines, podcasts).

Question: [5-13] What is cybercrime?

Answer: Cybercrime means doing illegal work using computers and the internet — such as hacking, phishing, cyberbullying, ransomware, malware, identity theft, software piracy, and distribution of illegal material. Those who do these acts are called cybercriminals.

Question: [5-14] What are the dangers of the internet?

Answer: Health dangers (too much screen use: lack of sleep, anxiety, depression, physical weakness), security dangers (identity theft, phishing, web attacks, spam, ransomware), social dangers (effects on social relations, relations with strangers), and effects on education (lack of attention, too much dependence on online resources).

Question: [5-15] Tell two ways to keep online information safe.

Answer: (1) Use a strong password — at least 12 characters, capital and small letters, numbers, and special symbols, and a separate password for each account. (2) Use two-factor authentication (2FA) — logging in needs a second code along with the password. Besides this, avoid sharing your personal information on the internet, and keep your software and apps updated.

Question: [5-16] Activity: make a strong password (10 minutes).

Answer: Rules for a strong password: at least 12 characters; include capital letters, small letters, numbers, and special symbols; no personal information (name, date of birth); avoid common words like abc123 or qwerty. The book's example: use a passphrase, on the pattern of "P@$$word2481" — a mix of letters, numbers, and symbols. Write your password in a safe place in your notebook and share it with no one.

Question: [5-17] Activity: write steps for facing cyberbullying.

Answer: (1) Check your school's anti-bullying policy. (2) Block the bully in the app. (3) Report the bully's contact. (4) Save screenshots as evidence. (5) Tell a trusted adult (teacher or parent) and report it. Instead of answering the bully, blocking and reporting is better.

Question: [5-18] Activity: check the privacy settings of your social media account (10 minutes).

Answer: Steps: (1) Log in to your account. (2) Go to Privacy in Settings. (3)

See who can see your posts and personal information. (4) Set the settings so that only friends can see your posts and personal information. (5) Limit any information that is open to everyone.

کارآفريني

[6-1] What is the basic purpose of a business plan?

(a) Showing products

✓ (b) Starting, operating, and growing a business

(c) Marketing without customers

(d) Only comparing prices

[6-2] Which of the following is NOT a part of a business plan?

(a) Market need

(b) Product design

✓ (c) Celebrity endorsement

(d) Marketing

[6-3] What does 'SEO' mean in digital marketing?

(a) Making a website more attractive

✓ (b) Improving search engine results

(c) Increasing sales options

(d) Ordering standard tools

[6-4] Which of the following is a modern payment method?

(a) Cash payment

(b) Credit card

✓ (c) Digital wallet

(d) Barter system

[6-5] Which of the following is commonly used for the digital marketing of products?

(a) Radio advertisements

(b) Billboards

✓ (c) Social media platforms

(d) Flyers

[6-6] Fill in the blanks: (i) Market research is necessary to understand customers' _____ and wants.

Answer: needs

[6-7] Fill in the blanks: (ii) A business plan makes clear a company's _____ and the ways to achieve them.

Answer: goals

[6-8] Fill in the blanks: (iii) Search Engine Optimization (SEO) helps to improve a website's _____ so that it can be found easily in search.

Answer: ranking

[6-9] Fill in the blanks: (iv) Digital marketing reaches more customers by using _____.

Answer: platforms such as social media.

[6-10] Fill in the blanks: (v) _____ carries out transactions in a safe way and keeps personal data secure.

Answer: Safe Transaction Method — a safe transaction method

Question: [6-11] What are the components of a business plan?

Answer: The important components of a business plan are: market need, customer need, product design, marketing (4Ps: Product, Price, Promotion, Place), costing, operations, and financial planning.

Question: [6-12] What is the difference between online and offline payment methods? How do you choose a safe method?

Answer: Offline methods: cash payment and credit cards — simple, but cash can be lost and card information can be stolen. Online methods: digital wallets, biometric verification, two-factor authentication (2FA), and smart contracts — fast and secure. To choose a safe method: use a well-known and trusted platform, turn on 2FA, share your information with no one, and check the receipt after payment.

Question: [6-13] Which tools are used in digital marketing?

Answer: Website designing, email marketing, social media marketing, online advertising, content marketing, mobile apps, and Search Engine Optimization (SEO).

Question: [6-14] What is a digital marketing plan?

Answer: A digital marketing plan decides how a business will sell its products or services through the internet — for example: the goal (how many customers to get), which platforms to use (Instagram, YouTube, website), what content to post, how much budget to keep, and how to measure success.

Question: [6-15] What is Search Engine Optimization (SEO), and how is it used with social media like Instagram?

Answer: SEO is the process by which a website comes up higher in search engine results. With Instagram: use the right words (keywords) in your posts, such as "fun cupcakes", include words in your profile and bio, post interesting pictures and videos, use hashtags, and put your website link in your bio — so that people can find you easily.

Question: [6-16] Teacher guidance: (1) Internet access on mobile. (2) The fun cupcakes example. (3) Visual aids. (4) SEO for a website. (5) Buying and selling role-play.

Answer: (1) Make sure students have internet access on a mobile device. (2) Use the "fun cupcakes" example to explain business and digital marketing. (3) Use visual aids (charts, pictures) to explain the steps of a business plan. (4) According to each topic, teach students to use SEO techniques for a website. (5) Include students in a buying-and-selling role-play — for example, a scene of buying and selling mobile accessories.